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$$\begin{aligned}f(x) &= 4\sqrt{x+1} \\&= 4(x+1)^{\frac{1}{2}} \\f'(x) &= 4 \cdot \frac{d}{dx} \left((x+1)^{\frac{1}{2}} \right) \\&= 4 \cdot \frac{1}{2}(x+1)^{-\frac{1}{2}} \cdot \frac{d}{dx}(x+1) \\&= 2(x+1)^{-\frac{1}{2}} \\f''(x) &= 2 \cdot \frac{d}{dx} \left((x+1)^{-\frac{1}{2}} \right) \\&= 2 \cdot \frac{-1}{2}(x+1)^{-\frac{3}{2}} \cdot \frac{d}{dx}(x+1) \\&= -(x+1)^{-\frac{3}{2}} \\&= -\frac{1}{\sqrt{(x+1)^3}}\end{aligned}$$

References